Problem(Farm animals) - Generate examples

Now that we have understood the problem statement, let us write down a few more sample cases or examples.
Review the sample input / output given in the table below.

Sample 1:

Input	Output
8 sample inputs	8 sample inputs
2 3 4	CHICKEN
2 3 5	NONE
2 3 6	ANY
2 3 7	NONE
2 3 8	CHICKEN
2 3 9	DUCK
2 3 10	CHICKEN
2 3 11	NONE

Did you like the problem?
85 users found this helpful

Select the correct options from the following.

- [x] For X = 3, Y = 4, Z = 12, The animal on the farm is 'ANY'.
- [ ] For X = 3, Y = 4, Z = 12, The animal on the farm is 'CHICKEN'.
- [ ] X = 3, Y = 5, Z = 13, The animal on the farm is 'CHICKEN'
- [ ] X = 3, Y = 5, Z = 13, The animal on the farm is 'NONE'
- [ ] X = 3, Y = 5, Z = 9, The animal on the farm is 'CHICKEN'
- [ ] X = 3, Y = 5, Z = 9, The animal on the farm is 'DUCK'

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Problem(Farm animals) - Solve sub-components

How do we find the following?

- Z is divisible only by X
- Z is divisible only by Y
- Z is divisible by both X and Y
- Z is divisible by neither X nor Y

Code it out in the IDE and click on **Submit** to continue.

Did you like the problem?

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Python3

```python
X = 3
Y = 4
Z = 12
if (Z%X == 0) and (Z%Y == 0):
    print('Z is divisible by both X and Y')
elif (Z%X == 0):
    print('Z is divisible only by X')
elif (Z%Y == 0):
    print('Z is divisible only by Y')
else:
    print('Z is divisible by neither X nor Y')

X = 3
Y = 5
Z = 13
if (Z%X == 0) and (Z%Y == 0):
    print('Z is divisible by both X and Y')
elif (Z%X == 0):
    print('Z is divisible only by X')
elif (Z%Y== 0):
    print('Z is divisible only by Y')
else:
    print('Z is divisible by neither X nor Y')
```

Your Output

```
Z is divisible by both X and Y
Z is divisible by neither X nor Y
```

Excellent work!

Submit Next

Problem(Farm animals) - Solve the complete problem

In Chefland, each chicken has X legs and each duck has Y legs.
Chef's farm can have **exactly one type** of bird.
Given that the birds on the farm have a total of Z legs:

- Print `CHICKEN`, if the farm can have only chickens but not ducks.
- Print `DUCK`, if the farm can have only ducks but not chickens.
- Print `ANY`, if the farm can have either chickens or ducks.
- Print `NONE`, if the farm can have neither chickens nor ducks.

Input format

- The first line will contain T, the number of test cases. Then the test cases follow.
- Each test case consists of a single line of input, containing three space-separated integers X, Y, Z.

Sample 1:

Input	Output
3 2 3 5 2 2 2 3 4 6	NONE ANY CHICKEN

Explanation:

Test case 1: There are 5 legs on the farm. Each chicken has 2 legs and each duck has 3 legs. Thus, the farm can have neither chickens nor ducks.

Test case 2: There are 2 legs on the farm. Each chicken has 2 legs and each duck has 2 legs. Thus, the farm can have any bird among chicken and duck.

Test case 3: There are 6 legs on the farm. Each chicken has 3 legs and each duck has 4 legs. Thus, the farm can have chicken only.

Did you like the problem?

26 users found this helpful

Python3

```python
# Update the program below to solve the problem

t = int(input())
for i in range(t):
    X, Y, Z = map(int, input().split())
    if (Z%X == 0) and (Z%Y == 0):
        print('ANY')
    elif (Z%X == 0):
        print('CHICKEN')
    elif (Z%Y == 0):
        print('DUCK')
    else:
        print('None')
```

Test against Custom Input

```
3
2 3 5
2 2 2
3 4 6
```

Sample Input

```
3
2 3 5
2 2 2
3 4 6
```

Your Output

```
None
ANY
CHICKEN
```

Run Submit Next

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Statement AI Help

Problem 3 (High Accuracy) - Generate examples

Now that we have understood the problem statement, let us write down a few more sample cases or examples.
Review the sample input / output given in the table below.

Sample 1:

Input	Output
6 sample inputs 11 12 13 14 15 16	6 expected outputs 1 0 2 1 0 2

Select the correct options from the following

- [x] If X = 18, minimum number of problems Chef marked incorrect are 0.
- [ ] If X = 18, minimum number of problems Chef marked incorrect are 2.
- [ ] If X = 20, minimum number of problems Chef marked incorrect are 2.
- [x] If X = 20, minimum number of problems Chef marked incorrect are 1.

✔ Hooray, you did it!

See Answer Hide Solution

Correct Answer(s):
If X = 18, minimum number of problems Chef marked incorrect are 0.
If X = 20, minimum number of problems Chef marked incorrect are 1.

Explanation:
If X = 18, minimum number of problems Chef marked incorrect are 0.
Chef will solve 6 questions correctly and score 18.
Min questions solved incorrectly is 0.

If X = 20, minimum number of problems Chef marked incorrect are 1.
If Chef solves 6 questions - his score will be 18.
Hence Chef will solve 7 questions - his score will be 21.
Hence to get a final score of 20 - Chef can get only 1 question incorrect.

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Problem 3 (High Accuracy) - Solve sub-components

How do we find the number of questions incorrectly solved?

- If X is divisible by 3, then minimum number of questions incorrect is 0
- If $X+1$ is divisible by 3, then minimum number of questions incorrect is 1
- If $X+2$ is divisible by 3, then minimum number of questions incorrect is 2

Go ahead and code out this sub-problem in the IDE!

```python
#Update the '_' below to solve the problem

X = 30
if X%3  == 0:
    print('Number of questions incorrectly solved is 0')
elif (X+1)%3 == 0:
    print('Number of questions incorrectly solved is 1')
else:
    print('Number of questions incorrectly solved is 2')

X = 34
if X%3 == 0:
    print('Number of questions incorrectly solved is 0')
elif (X+1)%3 == 0:
    print('Number of questions incorrectly solved is 1')
else:
    print('Number of questions incorrectly solved is 2')
```

Your Output

```
Number of questions incorrectly solved is 0
Number of questions incorrectly solved is 2
```

✔ Awesome, you nailed it!